

VSSV Aditya Ram

AI Engineer | GenAI, Azure & Backend Systems | Client-Facing Delivery
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PROFESSIONAL SUMMARY

AI Engineer at Tata Consultancy Services building and deploying LLM applications and AI-assisted operations systems for enterprise clients in beauty, telecommunications and roofing. Work hands-on with Azure OpenAI, Azure AI Foundry, Azure Functions, Cosmos DB, Azure AI Search, Ubuntu VMs and FastAPI. Design systems where the model reasons and deterministic code makes the decisions: tool allow-lists, grounding and citation validation, human approval gates, automated tests and measured latency. Present POCs, write technical briefings and run solution demonstrations for client stakeholders.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, SQL, Bash

GenAI and Agents: Azure OpenAI, Azure AI Foundry, GPT-5 / GPT-5-mini, Realtime API, WebRTC, LangGraph, LangChain, RAG, structured outputs / JSON Schema, embeddings and vector search, grounding, citation validation, confidence gating, human-in-the-loop, Gemini API, Ollama

Azure and Cloud: Azure Virtual Machines (Ubuntu), Azure Functions (Python v2), Azure Cosmos DB for NoSQL, Azure AI Search, Azure Network Security Groups, Application Insights, Cloudflare Workers / Vectorize

Backend and Software: FastAPI, REST / OpenAPI, Pydantic, httpx, pytest, Streamlit, Git, AI-assisted development (Claude Code, Codex)

ML and Data: PyTorch, Ultralytics YOLO, OpenCV, MediaPipe, scikit-learn, Pandas, NumPy, PyMuPDF, pdfplumber, Power BI

PROFESSIONAL EXPERIENCE

Systems Engineer - Data Science (AI/ML) | Tata Consultancy Services (TCS)

Hyderabad, India | Feb 2026 - Present

Incident Response AI - A global beauty leader

- Built an AI-assisted incident-diagnosis POC over 400+ existing monitoring rules; assembled 120-260-event evidence packs and used GPT-5 on Azure AI Foundry to correlate timelines, rank root-cause hypotheses with supporting/contradicting evidence, and propose recovery behind human approval.
- Engineered the AI gateway with citation validation, JSON-repair and 429 retries, a two-slot queue, grounding/approval checks and disk fallback; reduced live diagnosis latency from 124 s to ~70 s.
- Rebuilt the demo from three months of real operations exports with a deterministic Python pipeline and reproduced the client's Sonar dashboard in React, TypeScript and Tailwind; deployed on an Azure Ubuntu VM with nginx, systemd, TLS, auth and request limits; load-tested ~30 req/s for 90 s with zero errors at ~3% CPU.
- Present and demonstrate the solution to the client's operations stakeholders, supported by an AI business-case briefing (AI on top of the client's existing rules: understand, not detect) and a platform study guide.

Telecom customer assistant - Africa's largest mobile network operator | TCS Rapid Build team

- Built a FastAPI assistant with five Azure OpenAI call sites: GPT-5.4-nano routing, LangGraph multi-lookup planning, answer synthesis and gpt-realtime voice over WebRTC; enforced account access through a five-tool allow-listed registry so customer phone numbers never come from the model.
- Built core capabilities against a 22-feature requirements tracker, including Cosmos DB conversation persistence/session recovery, Azure AI Search conversation search, feedback capture, data cards and text-to-speech; mapped implementation and test status.
- Implemented mid-call English/Hindi/Telugu/Tamil switching, ephemeral Realtime tokens and a self-hosted coturn TURN relay over TLS/HMAC; built an Azure Function App (Python v2) with 13 HTTP routes and OpenAPI for Azure AI Foundry tool import.
- Owned Azure VM operations on a 1 GB Ubuntu host with nginx, certbot, systemd memory caps, NSG rules and repeatable redeloys; wrote 149 mocked unit tests and a 47-check live smoke suite; ~2.9 s typical chat latency.
- Evaluated four agent approaches, including Azure AI Foundry prompt agents, Microsoft Agent Framework and LangGraph; selected LangGraph for latency and control.

Customer-service and sales assistants - North America's largest roofing manufacturer

- As the sole engineer, designed and built two assistants end to end (Smart Order Helper and Product and Warranty Advisor): React frontend, FastAPI backend and Azure AI Foundry gpt-5-mini, with deterministic Python for pricing, discounts, inventory, credit and duplicate checks and human confirmation for every order.
- Extended the prototype into a sales-rep workflow: roof-size estimator, certified-contractor search, customer-inbox email agent, live WebRTC voice calls (Azure OpenAI Realtime) and a metrics tab, over a synthetic dataset of 24 products, 18 accounts and 164 contractors.
- Implemented a 95-point warranty auto-answer threshold and human review queue; fixed a hallucinated warranty term by moving evidence relevance into deterministic code and adding a regression test.
- Reduced gpt-5-mini latency from ~7-8 s to ~1.7 s, wrote 160 offline LLM/HTTP-mocked tests, deployed the backend on the shared Azure VM with a GitHub Pages frontend, and delivered the client demo.

Computer vision and internal delivery

- Trained a 14-class PPE detector with YOLO26n on a 44,000-image dataset; achieved mAP50 0.764, recall 0.80 and 6.3 ms inference, automated a 27-checkpoint sweep and profiled 80,398 labels for class imbalance.

Trainee Intern - Machine Learning | Tech Mahindra

Hyderabad, India | May 2024 - Aug 2024

- Developed IoT digital-twin prototype components spanning Bluetooth/TCP device communication, operator UI and a BLE Android companion app; built a Namaskar gesture-recognition model with OpenCV and MediaPipe.

Data Analyst Lead Intern | Ozibook

Remote | Aug 2023 - Oct 2023

- Led a 35-intern analytics team and built Power BI dashboards plus Selenium-based data-collection automation; reported 20% social-media engagement growth over the engagement period.

SELECTED TECHNICAL PROJECTS

Ask-about-Aditya - RAG assistant on Cloudflare

- TypeScript Worker with Cloudflare Vectorize and Workers AI embeddings; Gemini generation with Llama 3.3 70B fallback; grounded answers, citation ranking, origin-restricted CORS and secret-gated ingest.

Tata Passenger Vehicles Agentic OS - source-bound financial QA

- FastAPI/Streamlit pipeline over an FY25 annual report using PyMuPDF/pdfplumber, 12-topic chunking, text-embedding-3-large + FAISS retrieval, gpt-4o page-cited answers and a demo-mode guard.

Budget - invite-only finance PWA

- Cloudflare Workers application with PBKDF2-SHA256 authentication, HttpOnly sessions, 13-endpoint API and a confirmation-gated Gemini assistant; D1 database layer (8-migration schema) built with AI-assisted development (Claude Code, Codex).

PaperBrief AI - literature RAG and analysis

- Streamlit application using Ollama/phi3, embedding similarity, PyMuPDF and PubMed E-utilities for paper retrieval, comparison and knowledge reports; NetworkX used for relationship analysis.

EDUCATION

B.Tech, Computer Science and Engineering | GITAM University, Visakhapatnam | 2021 - 2025 | CGPA 8.11 / 10

CERTIFICATIONS & SELECTED SKILL BADGES

- Anthropic: Claude Certified Architect - Foundations; Claude Certified Developer - Foundations
- Google Cloud: Certified Partner Specialist - Gemini Enterprise Deployment; Certified Partner Specialist - Gemini Enterprise Agent Development
- Google Cloud skill badge: [Deploy an Agent with Agent Development Kit \(ADK\)](#)
- Google Cloud skill badge: [Evaluate and Improve Agent Development Kit Agents](#)
- Google Cloud skill badge: [Govern Agent Access with Gemini Enterprise Agent Platform](#)
- Google Cloud skill badge: [Develop Serverless Applications on Cloud Run Skill Badge](#)

ACHIEVEMENTS

- Runner-up, TCS Internal Hackathon (retail sentiment analysis with RAG); LIREL Champion, TCS; PwC Launchpad - top 5 on campus (Python and Salesforce programme)